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Short communication

Promoting strength-ductility synergy by heterogeneous grain structure in a zirconium-alloyed Fe-Mn-Al-C low-density steel

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ABSTRACT

Heterostructures, particularly dual-phase heterostructures, often lead to hetero-interfacial stress concentration, and ultimately premature failure, though they benefit from heterogeneous deformation. Addressing the challenge of strain localization and hetero-interfacial cracking, a single-phase heterogeneous grain structure is designed through tuning zirconium (Zr) content and employing precise thermo-mechanical treatment. As Zr content increases to 0.50 wt %, a single-phase bimodal grain structure is successfully engineered, including soft coarse grain zones (~5 μm in size and 65 % in area) and hard fine grain zones (~1 μm in size and 35 % in area). Such bimodal heterostructure is mainly ascribed to the retarding effect of recrystallization, due to the drag effect of Zr atoms and Zener pinning of ZrC. The synergistic effects of microstructural heterogeneity and ZrC precipitation enhance the strengths. Furthermore, multistage deformation mechanisms, involving heterogeneous deformation-induced hardening, dislocation slip and nanotwinning, enable the strain hardening capability and ductility.

1. Introduction

For structural alloys used in engineering applications, the simultaneous enhancement of strength and ductility remains a critical challenge. Recently, the Fe-Mn-Al-C-based steels have grained considerable attention, due to the desirable strength-ductility synergy and lightness [1–3]. To further enhance their mechanical properties, utilizing incoherent B2 intermetallic compounds has been a vital approach through addition of Ni [4,5]. However, such precipitation strengthening effect usually leads to a significant enhancement of strengths accompanied by a ductility loss, namely strength-ductility trade-off [6].

Recent development on alloys with heterostructues has a great potential to breakthrough such trade-off, thereby leading to various heterostructures introduced by different approaches including muti-phase structures [7,8], gradient grains [9], multi-modal grain structures [7–9]. For these heterostructures, the extra strengthening and hardening depends on the difference in hardness between the soft and hard regions, which is usually referred as heterogeneous deformation-induced (HDI) effect [7,8]. However, the pronounced disparity between soft and hard zones (e.g., hardness and elastic modulus) often induces the interfacial strain localization and premature failure, such as different dual-phase systems including FCC/B2 dual-phase steels [10], and FCC/$L2_1$ medium entropy alloys [11,12]. In comparison to these dual-phase alloys, heterogeneous grain structures usually exhibit hardness difference and an identical elastic modulus between soft and hard zones [13,14], owing to the single-phase structure. Such microstructural heterogeneities can effective mitigate strain localization via reducing disparity between hard and soft zones. Thus, introducing a single-phase heterogeneous grain structure is a promising pathway to enhancing the strength and ductility.

Following this line of thought, we introduce heterostructure grain structure via a combination of thermo-mechanical treatment (TMT) and addition of Zr. A typical single-phase heterostructure grain structure including coarse grains and fine grains is successfully designed via tuning Zr content. Such heterostructure effectively provides an extra strengthening and hardening effect, thereby further enhancing the strength and strain hardening capability.

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2. Materials and methods

2.1. Material preparation

Three Fe-Mn-Al-C low density steels were melted using a vacuum arc-melt furnace, and their chemical compositions are Fe-25Mn-7Al-0.9C-0Zr, Fe-25Mn-7Al-0.9C-0.01Zr and Fe-25Mn-7Al-0.9C-0.50Zr (wt. %), which are referred to as Zr0, Zr0.01 and Zr0.50 steels, respectively. For the melting process, the raw materials (>99.8 wt % pure) were placed into the water-cooled copper crucible. Then, the furnace was evacuated to a high vacuum level (the pressure below 10^{-6} bar) to avoid gas contamination including oxygen (O), nitrogen (N) and hydrogen (H). Additionally, introducing high-purity argon (Ar) into the furnace chamber (a pressure below 1 bar) prevents the oxidation and contamination. Repeating melting was conducted at least five times to further promote the uniformly chemical compositions and reduce the gas contamination. Such melting process including high-purity raw materials, a high vacuum level and high-purity argon can effectively achieve the average content of O (12–89 ppm [15]), N (~18 ppm [16]) and H (~2 ppm [17])during the steelmaking process. These as-cast steels were subjected to homogenization (1200 °C for 2 h), and Fig. S1 reveals that the oxygen segregate to grain boundaries does not occur. Then, hot-rolling with a thickness reduction of 40 %, followed by water quenching. These hot-rolled plates were subjected to cold rolling process with an 80 % reduction. Then, the Zr-alloyed cold-rolled plates were annealed at 900 °C for 5 min, followed by water-quenching.

2.2. Mechanical testing and materials characterization

For the room temperature tensile experiments, flat dog-bone-shaped samples with a gauge length of 14 mm and width of 4 mm were prepared from the cold-rolling plate along rolling direction (RD). The uniaxial tensile testing was conducted using Instron 5892, and the experimental data were recorded via a contact extensometer at a strain rate of 10^{-3} s^{-1}. To reveal the local strain evolution of different microstructures, a quasi-in-situ tensile testing with EBSD and μ-DIC analysis, was conducted. Specifically, the sample surface was prepared by mechanical grinding and polishing. Then, a monolayer of nanoscale SiO_2 particles was dispersed on the polished sample surface. The sample was then subjected to a quasi in-situ tensile deformation using the Instron 5892 universal testing machine. Further, the deformed sample surface with nanoscale SiO_2 was captured using a scanning electron microscope (SEM, Hitachi SU7000). These images were taken at the global strain level of 0 %, 2 % and 8 %, respectively, and then were analyzed using VIC-2D software to obtain local strain fields at different strain levels. Additionally, backscatter electron imaging (BSE) and electron backscatter diffraction (EBSD) analysis were carried out using SEM equipped with an EBSD detector (accelerating voltage: 30 kV, step size: 500 nm). For transmission electron microscopy (TEM) observations, TEM foils were prepared by twin-jet electro-polishing with a mixture of 5 % perchloric acid and 95 % ethanol (vol%). The TEM/high-resolution TEM (HRTEM) images and selected area electron diffraction (SAED) patterns were carried out by JEOL F200 operated at 200 kV.

3. Results

3.1. Microstructures of the Fe-Mn-Al-C low density steels with different Zr content prior to deformation

Fig. 1 shows the microstructure of the Fe-Mn-Al-C low density steel with different Zr content prior to tensile deformation. As shown in Fig. 1a, the microstructure of the Fe-Mn-Al-C low density steel without Zr is mainly composed of coarse recrystallized FCC grains, indicating that the steel annealed at 900 °C is subjected to full recrystallization. As adding Zr into the Fe-Mn-Al-C low density steel, Fig. 1b reveals that the microstructure of the Zr0.01 steel consists of coarse FCC grains and few fine FCC grains. With increasing Zr content to 0.50 wt %, the microstructure of the Zr0.50 steel exhibits a typical heterogeneous feature, i.e., coarse FCC grains (CG) and fine FCC grains (FG), as captured in Fig. 1c. An enlarged BSE image (Fig. 1c_1) clearly reveals that a high density of fine FCC particles is uniformly distributed in the FG regions.

To further reveal the microstructure evolution of the Fe-Mn-Al-C low density steels, the EBSD analysis was conducted and the results are shown in Fig. 2. For the Zr0 steel, the EBSD data including inverse pole figure (IPF), image quality (IQ) and kernel average misorientation (KAM) maps (Fig. 2a–a_2) reveal that its microstructure has equiaxed FCC grains with low KAM values. Further adding micro-alloyed Zr in the Fe-Mn-Al-C low density steel, the average size of recrystallized FCC grains is

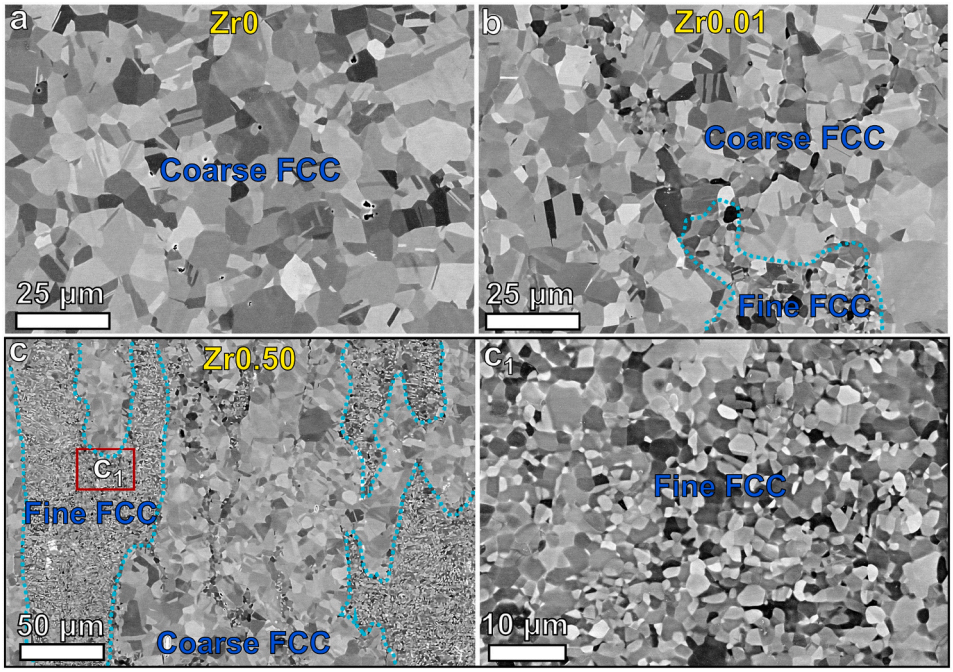


Fig. 1. BSE imaging of the Zr-alloyed Fe-Mn-Al-C low density steels prior to deformation. (a) Zr0 steel. (b) Zr0.01 steel. (c) and (c_1) Zr0.50 steel.

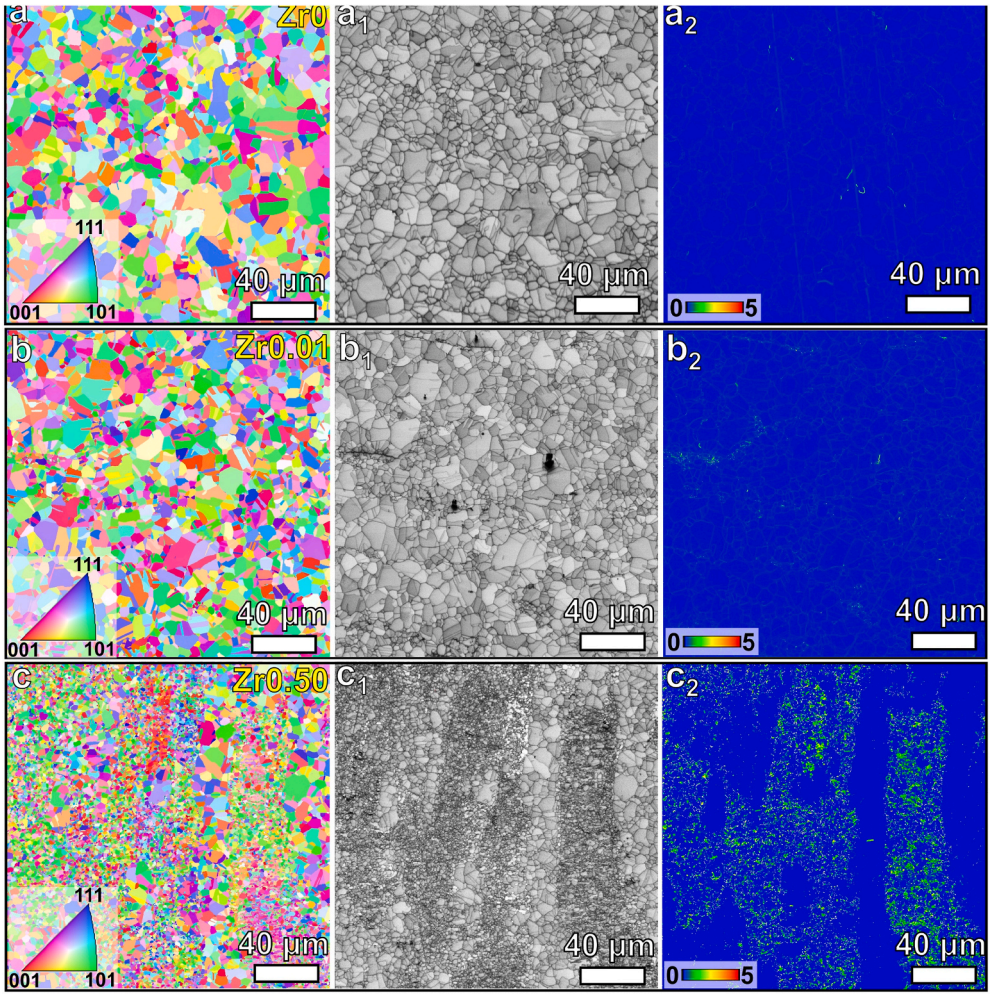


Fig. 2. EBSD data including IPF, IQ and KAM maps of the Zr-alloyed Fe-Mn-Al-C low density steels prior to deformation. (a)-(a_2) Zr0 steel. (b)-(b_2) Zr0.01 steel. (c)-(c_2) Zr0.50 steel.

obviously reduced and their grain orientations are also random, as shown in Fig. 2b–b_2. Differently, Fig. 2c–c_2 reveals that the microstructure exhibits a distinct heterostructure including CG zones and FG zones. Compared to the CG zones with low KAM values, the FG zones exhibit a high KAM value, indicating a high density of dislocations are decorated in these fine FCC grains.

According to the BSE images and EBSD data, more microstructural details of CG zones and FG zones, including their average sizes and area fraction, are estimated and shown in Fig. 3. As shown in Fig. 3a, the fraction of fine grains is significantly increased (0 % → 35 %), as increasing Zr content from 0 wt% to 0.50 wt %. For the average size of FCC grains (Fig. 3b–d), the average size of FCC grains is obviously reduced (12 μm → 7 μm) when the steel is added to 0.01 wt % Zr. Further adding Zr into the steel (i.e., Zr0.50 steel), the FCC grains exhibit a bimodal distribution of 5 μm and 1 μm, resulting from the heterogeneous structure including CG zones and FG zones, which stands in contrast to the homogeneous FCC grains in Zr0 steel and Zr0.01 steel. These results suggest that increasing Zr content can lead to a heterogeneous architecture of FCC matrix.

3.2. Tensile deformation behavior

Fig. 4a shows the engineering stress-strain curves of the Fe-Mn-Al-C low density steels with different Zr content. In Zr0 steel, its yield strength is ~435 MPa, the ultimate tensile strength is 820 MPa, and fracture ductility is 52 %. As Zr content increased to 0.01 wt %, the strengths of the Zr0.01 steel are slightly increased, i.e., yield strength: ~450 MPa, ultimate tensile strength: ~832 MPa, while its fracture ductility is increased to 55 %. With further increasing of Zr content, the Zr0.50 steel exhibits a reasonable strength-ductility combination, i.e., yield strength: ~505 MPa, ultimate tensile strength: ~855 MPa and fracture ductility: 58 %. Similarly, a high true ultimate tensile strength and true strain is observed in the Zr0.50 steel. This suggests that introducing heterostructure in Fe-Mn-Al-C low-density steels through addition of Zr can enhance the mechanical properties. The microscopic fracture morphologies of these fractured Fe-Mn-Al-C low density steels are shown in Fig. 4b–d. For the Zr0 steel, its facture surface is mainly composed of small dimples and cleavage faces. As the Zr content increasing, the dimples are obviously larger and deeper than those in Zr0 steel, and the density is also increased.

3.3. Deformation behviors after tensile deformation

To reveal the strain partitioning behavior in the hetero-structured Zr0.50 steel, the von Mises maps are obtained from quasi in-situ μ-DIC as a function of global strain in Fig. 5. The EBSD-IQ map (Fig. 5a) shows the measured area comprising the coarse FCC grains and fine FCC grains. As illustrated in Fig. 5b, significant strain partitioning is observed at a minor plastic strain of 2 %. The CG zones and hetero-interfaces between FG and CG zones experiences high strain, whereas the FG zones remain nearly undeformed. This is mainly attributed to the hardness differences between CG zones (soft regions) and FG zones (hard regions). As the

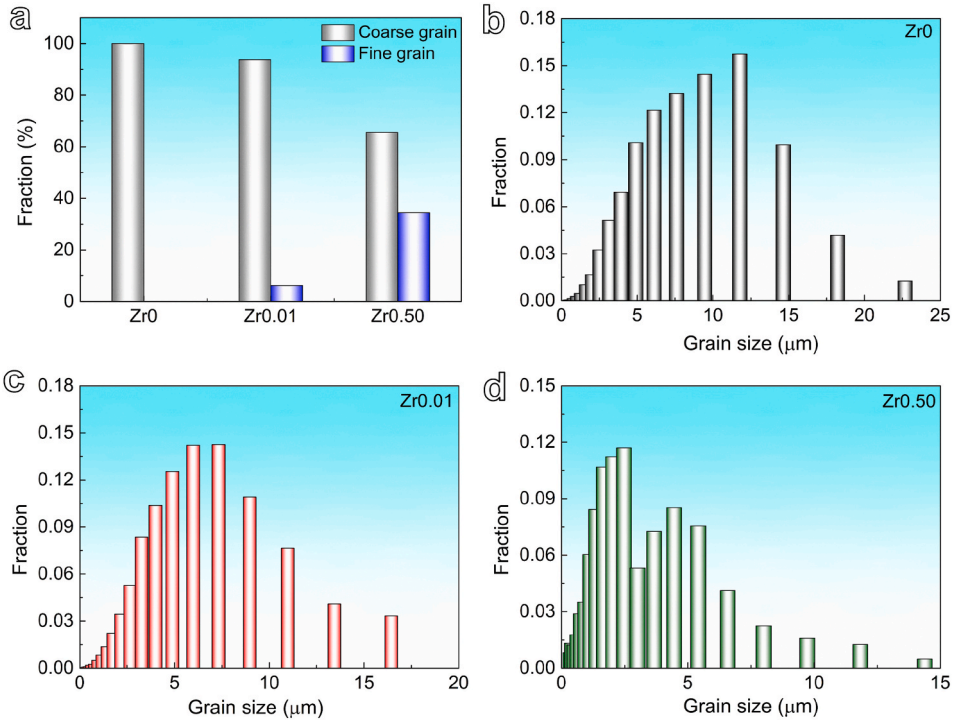


Fig. 3. (a) Area fraction of FG and CG zones in Zr0, Zr0.01 and Zr0.50 steels. (b)–(d) Grain size distribution of FCC grains in Zr0, Zr0.01 and Zr0.50 steels, respectively.

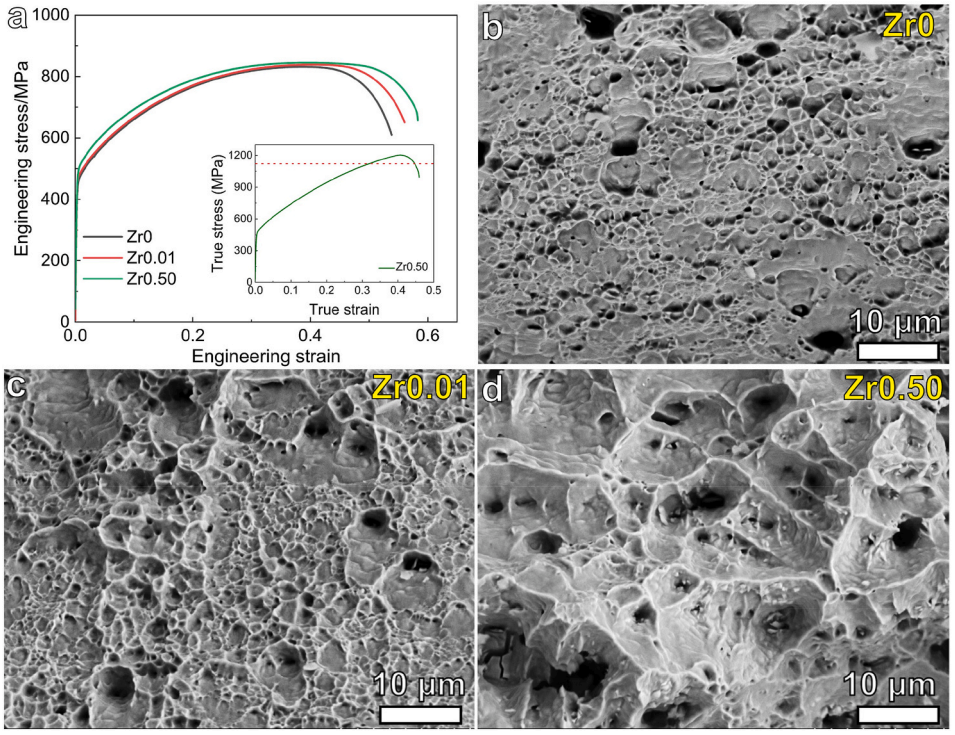


Fig. 4. (a) Engineering stress-strain curves of the Zr-alloyed Fe-Mn-Al-C lightweight steels, and the inset is the true stress-strain curve of the Zr0.50 steel. (b)–(d) Fracture surfaces of Zr0, Zr0.01 and Zr0.50 steels, respectively.

global strain increased to 8 % (Fig. 5c), the local strain of the FG zones becomes more pronounced, indicating that the strain partitioning between soft and hard regions persists during plastic deformation. Such strain partitioning behavior leads to the generation of geometrically necessary dislocations (GNDs), promoting a heterogeneous deformation-induced (HDI) strengthening [7,8].

To grain deeper insights into the deformation mechanisms of the Zr0.50 steel, TEM/HRTEM based techniques were carried out. Fig. 6a shows that dislocations as a vital deformation mechanism are initially activated in FCC matrix during a minor of plastic deformation. At the

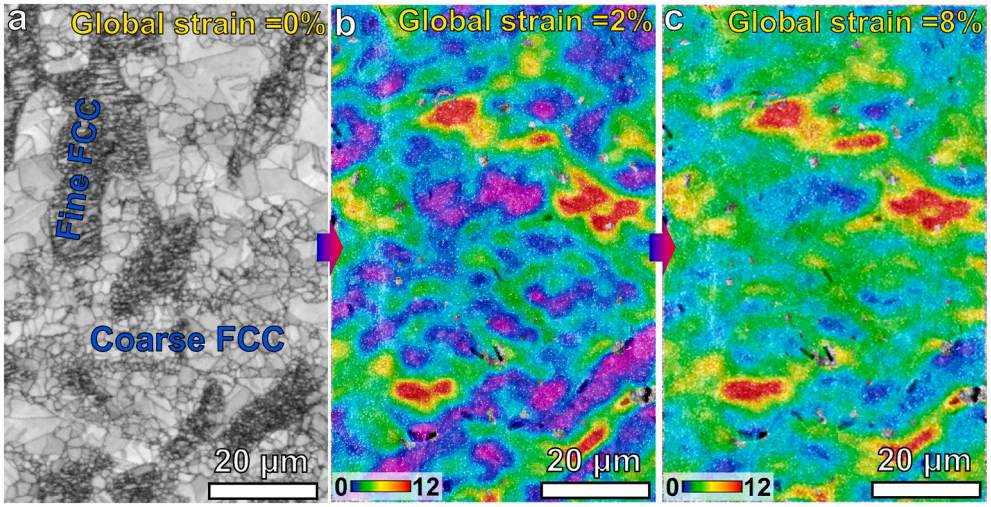


Fig. 5. (a) EBSD-IQ map of the Zr0.50 steel without global strain showing FG and CG zones. (b) and (c) Von Mises strain maps at 2 % and 8 % global strain, respectively, revealing the strain partitioning behavior between FG and CG zones.

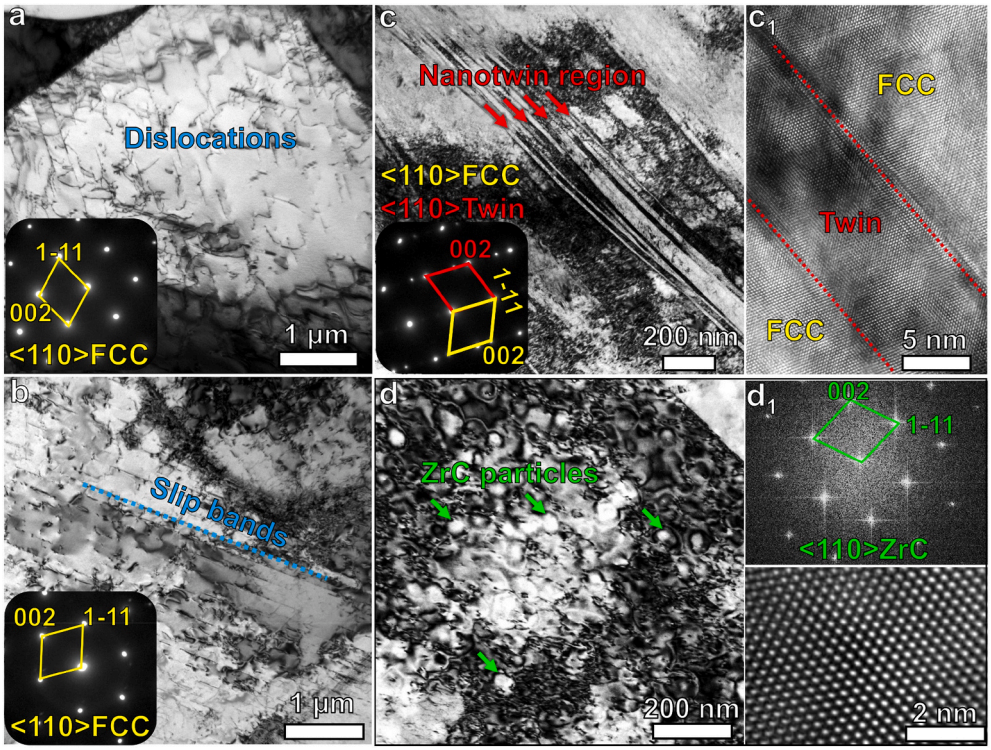


Fig. 6. Microstructure characterization of the Zr0.50 steel during tensile deformation. (a) TEM imaging shows a planar slip in FCC grain interior after a minor deformation. (b) TEM imaging reveals the slip bands after a severe deformation. (c) TEM imaging displaying a high density of nanotwins (indicated by red arrows). (c_1) HRTEM imaging showing a nanoscale nanotwin. (d) TEM imaging showing the interaction between dislocations and nanoscale ZrC particles. (d_1) FFT pattern and HRTEM image confirm the ZrC carbide. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

fracture surface, the severe deformed FCC matrix (Fig. 6b) exhibits a high density of slip bands along 111planes, which generally leads to a strengthening effect [18]. Excepted for dislocations, nanotwins with ~20 nm in size are generated during a heavy deformation, as evidenced by a TEM imaging (Fig. 6c). In the meantime, a high density of dislocation is decorated around the nanotwinned region, which contributes to the interaction between dislocations and nanotwins. Such behavior further effectively reduces the mean free paths of mobile dislocations, thereby leading to a typical twinning-induced plasticity. Fig. 6c_1 reveals that twin boundaries are decorated by multiple stacking faults, which indicates that the nanotwins formation originates from the transition from multiple Shockley partial dislocations to twins [19]. In addition to the deformation behaviors of FCC matrix, Fig. 6d shows that numerous particles with ~30 nm in size are distributed in FCC grain interior, which can effectively inhibit the movement of dislocations, i.e., precipitation strengthening. A detail analysis on these nanoscale particles via HRTEM and corresponding FFT pattern (Fig. 6d_1) confirms that these nanosized particles are identified as ZrC carbides, due to the addition of Zr and the strong affinity between Zr and C [20].

4. Discussion

4.1. Formation mechanism of the heterogeneous grain structure

In general, heterogeneous grain structures, e.g., bimodal recrystallized grains (CG/FG zones) [9], recrystallized and non-recrystallized zones [13], are designed through introducing partial recrystallization. The TMT, i.e., severe cold-rolling and various annealing conditions, is a significant approach for the partial recrystallization. In present work, a typical TMT treatment including 80 % cold-rolling and short-time annealing (900 °C for 5 min) was applied in the Fe-Mn-Al-C steels. However, the Zr0 steel exhibits a homogeneous grain structure after the TMT treatment. Interestingly, the combination of TMT treatment and Zr addition leads to a heterogeneous grain structure including CG and FG zones. As increasing Zr content, the fraction of FG zones is obviously increased (Fig. 3). Thus, it is necessary to reveal the formation mechanism of heterogeneous grain structure in Zr0.50 steel.

It is well accepted that recrystallization is driven by the stored energy of cold-rolled deformation, i.e., the high density of dislocations and shear bands. Such driving force (F_{rec}) is decreased by the retarding forces including solute drag (P_s) and Zener pinning (P_z), which is expressed by Ref. [21]:

$$F_{rec} = -\frac{\gamma_{gr}}{R_{rec}} + \frac{\gamma_{sub}}{R_{sub}} + c_d G b^2 \left(\rho_{def} - \rho_{rec}\right) - P_s - P_z \quad (1)$$

where γ_{gr} and γ_{sub} are boundary energy of subgrains and boundary energy of a high angle grain boundary, R_{rec} and R_{sub} are sizes of recrystallized grains and subgrains, c_d is a constant (0.5), ρ_{def} and ρ_{rec} are the dislocation densities in the deformed and recrystallized grains, G is the shear modulus, b is Burger's vector.

For the annealed Zr0.50 steel, Zr elements are mainly distributed in FCC matrix, except for precipitation of ZrC. Based on Equation (1), Zr in solid solution and ZrC precipitation exhibit a retarding effect of recrystallization. Specifically, these Zr elements distributed in FCC matrix serve as solute atoms, which usually diffuses to the vicinity of defects including grain boundaries, shear bands and dislocations. During annealing process, the accumulation of these Zr atoms in turn impedes the movement of grain boundaries and recovery of shear bands and dislocations. Such drag effect from Zr atoms leads to a high energy barrier of recrystallization of cold-rolled microstructure during annealing [11,22], thereby inducing the partial recrystallization. Further, according to the thermodynamic calculation (Fig. 7), the fraction of ZrC particles is ~0.6 % after annealing at 900 °C. These nanoscale ZrC particles also inhibit the migration of grain boundaries during recrystallization through a typical Zener pinning effect. The Zener pinning force is usually related to the average size of precipitates and fraction of precipitates [23]. Compared to the Zr0 and Zr0.01 steels, the fraction of ZrC carbides in Zr0.50 steel is highest. Based on Fig. 6d, the size of ZrC in the Zr0.50 steel is only 30 nm. As a result, these nanoscale ZrC carbides in the Zr0.50 steel can lead to a desirable Zener pinning force, thereby playing a pivotal role of the partial recrystallization. Additionally, the micro shear bands caused by the heavy cold-rolling deformation are the preferred sites for the nucleation of recrystallization [24,25], which provides the enough driving force to formation of fine grains. As a result, the synergetic effects of drag effect of Zr atoms, Zener pinning of ZrC and TMT treatment promote the heterogeneous grain structure in the Zr0.50 steel.

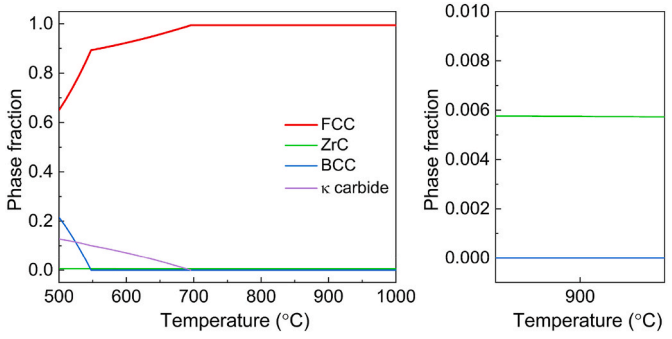


Fig. 7. Thermodynamic phase fraction calculation for the Zr0.50 steel using the Thermo-Calc software.

4.2. Effect of heterogeneous grain structure on enhancing strength-ductility synergy

The tensile results show that the Zr0.50 steel exhibits a desirable mechanical property compared to the Zr0 and Zr0.01 steels. For the yield strength, the bimodal heterogeneous grain structure induces an extra strengthening contribution (i.e., HDI strengthening), apart from traditional strengthening mechanisms, e.g., solid-solution strengthening, grain boundary strengthening and precipitation strengthening. In Zr0.50 steel with a bimodal grain structure (FG and CG zones), the contributions for yield strength including grain boundary strengthening and HDI strengthening can be expressed as [9]:

$$\sigma_{GB-HDI} = f_{FG}\sigma_{FG} + \left(f_{CG} - f_{CG,b}\right)\sigma_{CG} + f_{CG,b}\sigma_{CG,b} \quad (2)$$

where f_{FG} and f_{CG} are fractions of FGs and CGs, respectively, $f_{CG,b}$ is the fraction of those CGs adjacent to FGs, which is ~37 % based on EBSD result (Fig. 2). σ_{FG} and σ_{CG} can be estimated using Hall-Petch equation [26]:

$$\sigma = \sigma_0 + kd^{-1/2} \quad (3)$$

where σ_0 is the fraction stress (~97 MPa [10]), k values for coarse and fine FCC grains are 14.55 MPa $mm^{1/2}$ and 9.1 MPa $mm^{1/2}$, respectively [10].

$\sigma_{CG,\ b}$ is the back stress strengthened CGs adjacent to FGs, which is expressed as [9]:

$$\sigma_{CG,b} = \sigma_0 + \sqrt{\frac{2k^2}{d_{CG}d_{FG}}l_{Hbar} + \sigma_{CG}^2} \quad (4)$$

where l_{Hbar} is the width of hetero-boundary affected region, which is calculated by $\left[1 - (\sigma_{CG} - \sigma_{FG})^2\right]d_{CG}$. Based on Fig. 3, the l_{Hbar} is estimated to be 2.45 μm. Based on Equations (2)–(4), the strengthening contributions from grain boundary and HDI are ~300 MPa.

In present study, the precipitation of nanoscale ZrC carbides can promote the Orowan precipitation strengthening [27,28]:

$$\sigma_p = \left(\frac{0.538Gbf^{1/2}}{a}\right)\ln\left(\frac{a}{2b}\right) \quad (5)$$

Where f is the fraction of precipitates and a is the average size of precipitates. Based on Figs. 6 and 7, the average size and fraction of ZrC carbides are 30 nm and 0.006, respectively. The precipitation strengthening of ZrC carbides is calculated to be 108 MPa. As a result, the high yield strength in the Zr0.50 steel originates from grain boundary strengthening, HDI strengthening and precipitation strengthening.

Except for the enhanced yield strength, the Zr0.50 steel exhibits a remarkable strain hardening capability and fracture ductility, as shown in Fig. 4a. For the heterogeneous grain structure, a high density of GNDs is generated during the elasto-plastic co-deformation, as evidenced in Fig. 5b. As the global strain increasing, the co-deformation of soft CG zones and hard FG zones occurs, which further promotes the compatible deformation and induces additional strain hardening, which is expressed as [29]:

$$\frac{d\tau}{d\varepsilon}=\alpha Gb\frac{d\left(\sqrt{\rho_s+\rho_G}\right)}{d\varepsilon} \tag{6}$$

$$\frac{d\tau}{d\varepsilon}=\left(\frac{d\tau}{d\varepsilon}\right)_{dis}+\left(\frac{d\tau}{d\varepsilon}\right)_{HDI} \tag{7}$$

where ε and τ are strain and shear stress, respectively. ρ_S and ρ_G are the density of the statistically stored dislocations (SSDs) and GNDs, respectively. According to Equations (6) and (7), the density of GNDs is increased in the Zr0.50 steel with heterogeneous grain structures, which in turn assist an extra hardening capability, i.e., HDI hardening effect, and thereby leads to manifestation of HDI ductility [8].

Further, a high density of deformation twins can be successfully introduced during tensile deformation, apart from the dislocations. It is reported that formation of deformation twinning is related to the critical stress for twinning. The critical stress for twining in FCC-structured alloys is described as [29,30]:

$$\sigma_{tw}=M\left(\frac{SFE}{b}+\frac{Gb}{D}\right) \tag{8}$$

where M is the Taylor factor (3.06), SFE is the stacking faults energy, b is Burgers' vector (0.147 nm) and D is the grain size (Fig. 3). In present work, the SFE of FCC matrix in Zr0.50 steel is estimated to be ~55 mJ/m^2 (**Supplementary text S1**). Thus, the critical twinning stress of the Zr0.50 steel is ~1.14 GPa, which is obviously lower than its true ultimate tensile strength (the inset in Fig. 4a). Such high strengthening effect promotes the formation of deformation twinning, thereby enabling the further strain hardening capability. Thus, the synergic effects of HDI hardening and deformation twins provide a reasonable strain hardening rate, which further improves the fracture ductility.

5. Conclusion

In present study, a single-phase heterogeneous grain structure, containing coarse grains with ~5 μm in size and 65 % in area and fine grains with ~1 μm in size and 35 % in area, is successfully engineered in a Fe-Mn-Al-C low density steel. As the Zr content increases to 0.50 wt %, the drag effect of Zr atoms and Zener pinning of ZrC carbides is obvious enhanced, which can retard the recrystallization process. Such retarding recrystallization in turn assists the formation of bimodal grain structure. The heterogeneous grain structure has yielded desirable enhancement in strength-ductility synergy, in comparison to the Fe-Mn-Al-C steels without Zr addition or with 0.01 wt % Zr addition. The significant strengthening effect is mainly attributed to the grain boundary strengthening, HDI strengthening and precipitation strengthening. For the enhanced strain hardening capability and fracture ductility, the deformation mechanisms are dominated by cooperating slip bands, HDI hardening and deformation twins.

CRediT authorship contribution statement

Yuheng Wang: Writing – original draft, Investigation. **Runze Yu:** Writing – review & editing, Conceptualization. **Fengchao An:** Writing – review & editing, Funding acquisition. **Suotao Wang:** Writing – review & editing, Funding acquisition. **Hao Wang:** Resources, Methodology, Investigation. **Zibo Zhao:** Methodology, Formal analysis. **Hongpu Yue:** Resources, Methodology. **Tianxiang Gao:** Methodology, Investigation. **Junting Luo:** Writing – review & editing. **Junsong Zhang:** Writing – review & editing, Data curation. **Riping Liu:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.vacuum.2026.115094.

Data availability

Data will be made available on request.

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